HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Built and fine-tuned **NLP and ML models** using Python (scikit-learn, PyTorch, Hugging Face Transformers) for classification, entity extraction, and text generation tasks.
- Designed **end-to-end ML training and inference pipelines** with data preprocessing, feature engineering, model evaluation, and versioning using MLflow.
- Developed **LLM-powered automation scripts** using OpenAI and Hugging Face APIs, reducing manual document processing time by 40% for internal operations teams.
- Built **REST APIs with FastAPI and Flask** to serve ML model predictions as real-time microservices, integrated into Java Spring Boot-based backend systems.
- Worked with **PostgreSQL and AWS S3** to manage training datasets, feature stores, and model artefacts at scale, ensuring data lineage and reproducibility.
- Collaborated with product and data engineering teams in **Agile/Scrum** sprints to define ML problem framing, evaluation metrics, and production deployment criteria.
- Automated model retraining and evaluation workflows using **Jenkins and Python**, ensuring continuously updated models without manual pipeline intervention.
- Applied **prompt engineering and RAG (Retrieval Augmented Generation)** patterns to ground LLM outputs in domain-specific knowledge bases, improving answer accuracy by 35%.

SKILLS

Machine Learning & AI – scikit-learn, PyTorch, Hugging Face Transformers, LLM APIs, RAG, NLP.

Programming Languages – Python (advanced), SQL, Java, Bash scripting.

Python Ecosystem – pandas, NumPy, Flask, FastAPI, LangChain, MLflow.

Databases & Storage – PostgreSQL, MySQL, AWS S3, vector databases (Pinecone basics).

Cloud & MLOps – AWS (EC2, S3, Lambda, SageMaker basics), Docker, Jenkins, CI/CD pipelines.

Development Practices – Agile/Scrum, Git, experiment tracking, model versioning, prompt engineering.

Tools – Jupyter Notebook, VS Code, Postman, Jira.

PROJECTS

LLM-Powered Document Classification & Extraction (HCL Technologies)

- Built an NLP pipeline using Hugging Face Transformers to classify and extract structured data from unstructured support documents, reducing manual classification effort by 40%.
- Applied prompt engineering with OpenAI API to generate structured summaries from long-form technical reports, improving information retrieval speed for operations teams.
- Deployed the pipeline as a FastAPI microservice integrated into the existing Java Spring Boot backend, enabling real-time document processing.

ML-Driven Anomaly Detection for Operational Monitoring (HCL Technologies)

- Developed an anomaly detection system using scikit-learn (Isolation Forest, LSTM) to identify system metric deviations from historical baselines, reducing alert noise by 35%.
- Automated model retraining with Jenkins pipelines and stored artefacts in AWS S3, ensuring continuously updated detection thresholds without manual intervention.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)